STROKE

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Definition of Stroke

- ***Stroke*** (Cerebrovascular accident, CVA): rapidly developing clinical signs of focal or global disturbance of cerebral function, with symptoms lasting 24 hours or longer, or leading to death, with no apparent cause other than a vascular origin.......WHO, 1976

- Stroke definition by time course:
  - **Transient ischemia attack (TIA):** ischemic events $\leq$ 24 hours without apparent permanent neurological deficits
  - **Stoke in evolution:** progressive neurological deficits over time suggesting a widening of the area of ischemia
  - **Completed stroke:** ischemic event with persisted deficit

Epidemiology & Global Burden:

- Stroke is the **second leading cause of death** and a major cause of long-term disability worldwide.
- According to the **World Health Organization (WHO)**, approximately **15 million people** suffer a stroke each year, with **5 million deaths** and another **5 million left permanently disabled**.
- Stroke requires rapid diagnosis and intervention to minimize brain damage.
- The concept of "Time is Brain" emphasizes the importance of early thrombolysis (IV tPA) and mechanical thrombectomy in ischemic stroke.
- For hemorrhagic stroke, blood pressure control, surgical intervention, and neurocritical care are crucial.

Current concepts

- **TIA**: is defined as a transient episode of neurologic dysfunction caused by focal brain, spinal cord, or retinal ischemia, without acute infarction.
- **Stroke** : objective evidence of infarction irrespective of duration of clinical symptoms. CT/MRI necessary to increase diagnostic accuracy.

  (cf Angina pectoris vs Myocardial infarction).
- **CITS** Cerebral Infarction with Transient Sign
- **CIND** Cerebral Infarction No Deficit, silent

Stroke: Well-Documented and Modifiable Risk Factors

- **Hypertension**
- **Diabetes**
- **Dyslipidemia**
- Atrial fibrillation
- Other cardiac conditions
- Cigarette smoke
- Asymptomatic carotid stenosis
- Sickle cell disease
- Postmenopausal hormone therapy
- Diet and nutrition
- Physical Inactivity
- Obesity and body fat distribution

Stroke: Non-modifiable Risk factors

- Age
- Sex
- Ethnicity
- Prior stroke
- Heredity

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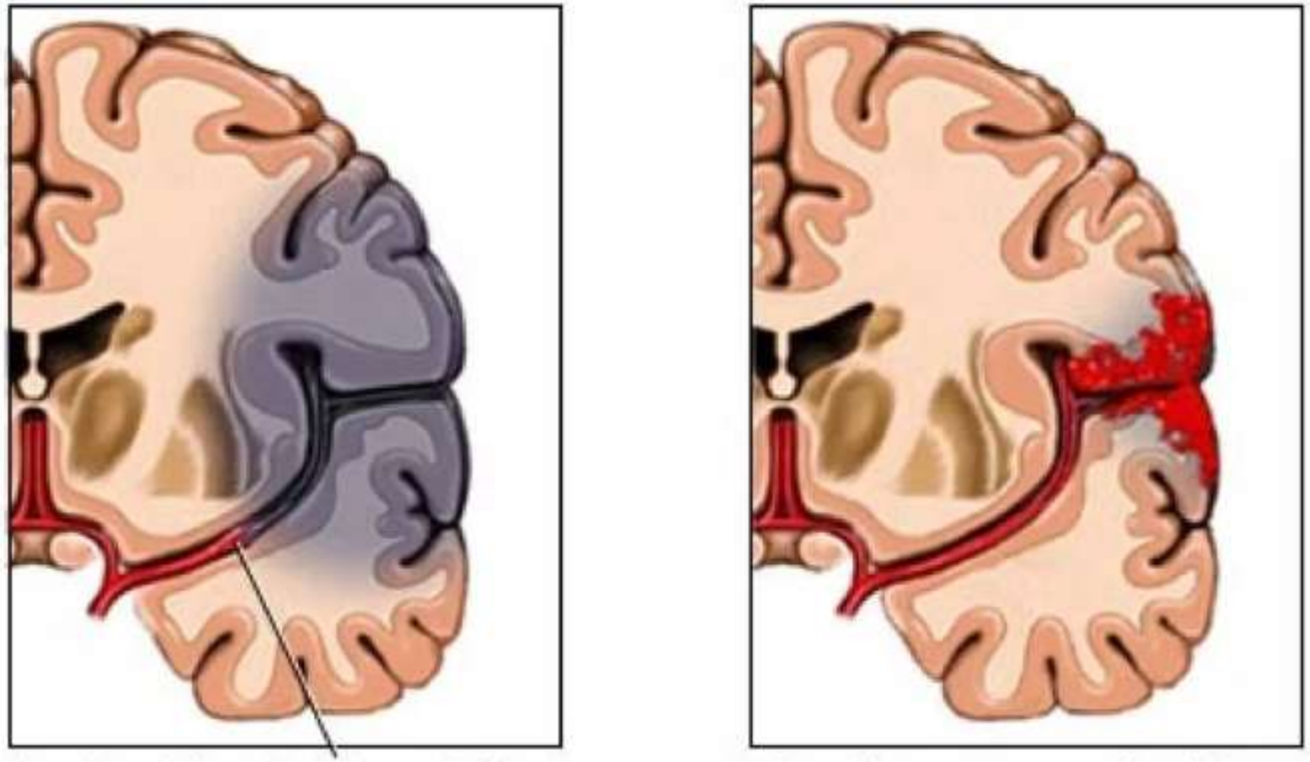
CLASSIFICATION OF STROKE

- **Stroke**
  - **Primary Ischemic (80% of Strokes)**
    - **Thrombotic** 50%
    - **Embolic** 30%
  - **Primary Hemorrhagic (20% of Strokes)**
    - **Intracerebral Hemorrhage 15%**
    - **Subarachnoid Hemorrhage 5%**

Two Major Types of Stroke

Ischemic stroke

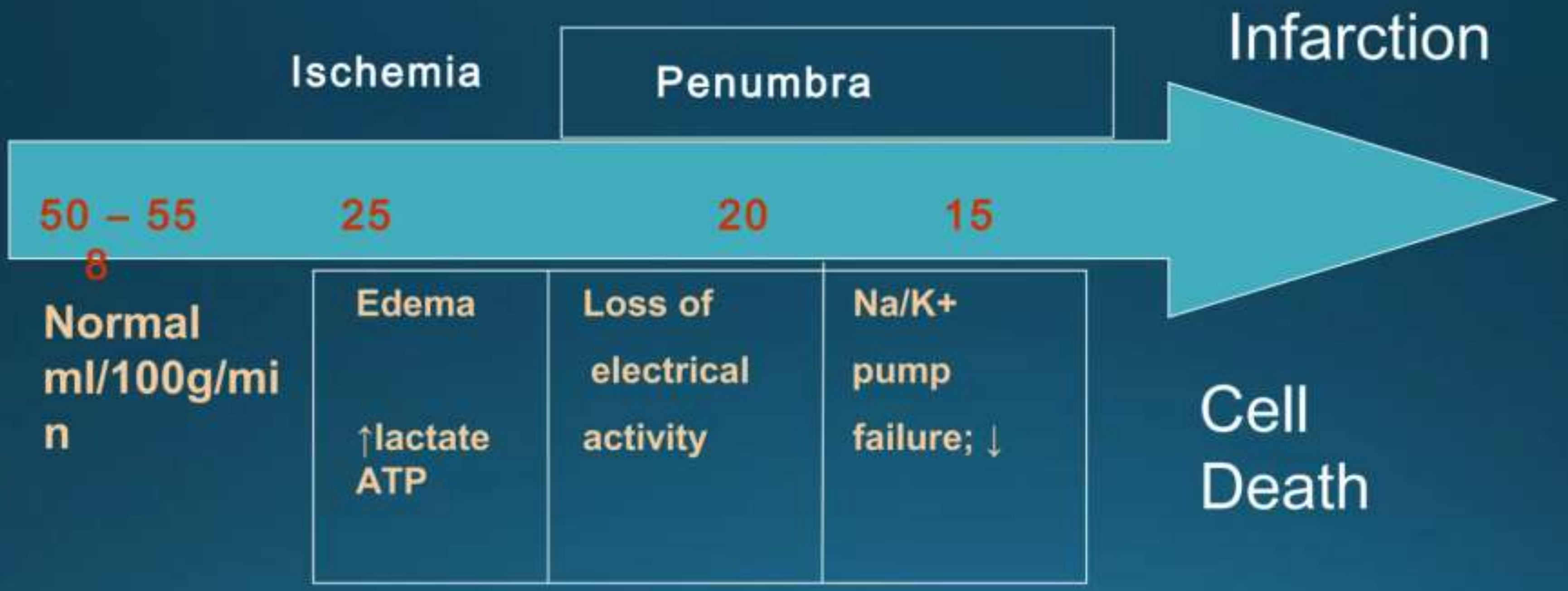
Hemorrhagic stroke



A clot blocks blood flow to an area of the brain

Bleeding occurs inside or around brain tissue

Effects of Reduced CBF



Pathophysiology of Stroke

- I. **Ischemic Stroke Pathophysiology**

1. **Atherosclerosis and Thombosis:**
   1. *Chronic hypertension, diabetes, smoking, and hyperlipidemia cause endothelial damage and atherosclerotic plaque formation.*
   2. *Plaque rupture triggers platelet aggregation and thrombus formation, leading to vessel occlusion.*
2. **Embolic Stroke:**
   1. *An embolus from the heart (e.g., atrial fibrillation, infective endocarditis) travels and blocks a cerebral artery.*
3. **Lacunar Stroke (Small Vessel Disease):**
   1. *Hypertension causes lipohyalinosis (thickening of vessel walls), leading to occlusion of small penetrating arteries.*

II. Hemorrhagic Stroke Pathophysiology

1. **Hypertensive Bleed (Intracerebral Hemorrhage - ICH)**
    1. *Chronic hypertension weakens small arteries, leading to rupture and hemorrhage.*
    2. *Hematoma formation causes increased intracranial pressure (ICP) and brain herniation.*
2. **Aneurysm Rupture (Subarachnoid Hemorrhage - SAH)**
    1. *Weakened blood vessel walls (Berry aneurysm) rupture due to hypertension or sudden stress.*
    2. *Blood in the subarachnoid space leads to vasospasm, ischemia, and hydrocephalus.*

Pathophysiology of Ischemic Brain Injury

Topography of focal ischemia

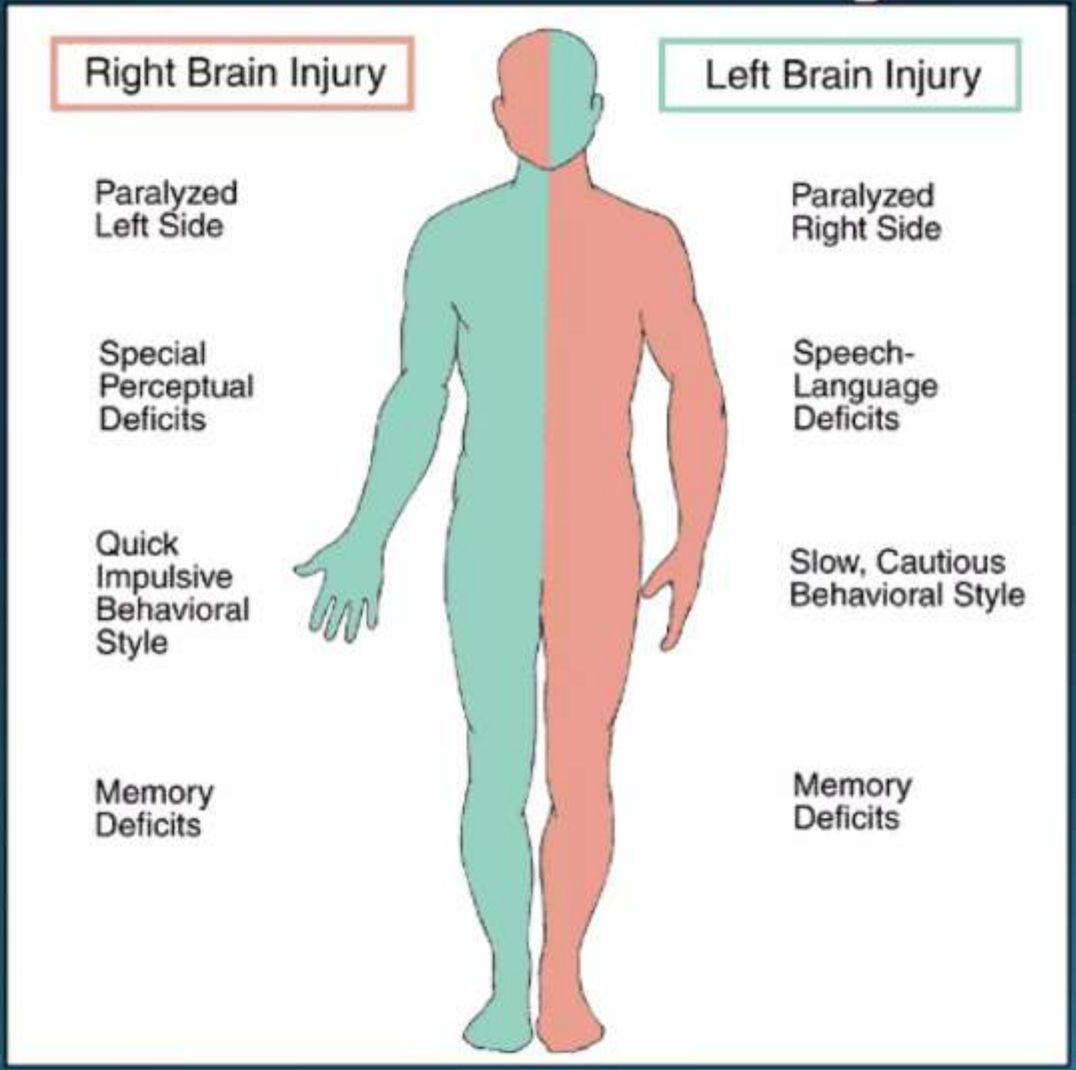
- Flow gradient: heterogeneous regional CBF reduction after focal ischemia
- Densely ischemia region surrounded by areas of less severe CBF reduction
- **Ischemic penumbra:** an area of reduced perfusion sufficient to cause potentially reversible clinical deficits but insufficient to cause disrupted ionic homeostasis

CLINICAL FEATURES OF STROKE

Stroke Warning Signs

- Sudden weakness or numbness of the face, arm or leg, especially on one side of the body
- Sudden confusion, trouble speaking or understanding
- Sudden trouble seeing in one or both eyes
- Sudden trouble walking, dizziness/vertigo, loss of balance or coordination
- Sudden, severe headaches with no known cause (for hemorrhagic stroke)

The symptoms of a stroke are dependant on what portion of the brain is damage.



INVESTIGATIONS

- full blood count, serum electrolytes, renal function tests, cardiac enzymes, and coagulation studies
- Blood sugar is mandatory to exclude hypoglycemia or diagnose diabetes mellitus
- Full blood count to detect Polycythaemia,ESR for endocarditis,
- clotting studies for Hypercoagulable States
- An electrocardiogram (ECG) : arrhythmias and myocardial infarction. Baseline ECG is recommended in all patients with stroke(AHA/ASA Guidelines)
- Echocardiography : valve disease and intra-cardiac clot

NEUROIMAGING

- Brain CT scan: CT is sensitive to the intracranial blood and is readily available.

Normal early CT therefore rules out haemorrhagic stroke. CT Scan changes in ischemic stroke may take several days to develop.

- MRI: MRI is better at detecting posterior fossa lesions especially in posterior circulation stroke such as Pons or cerebellum
- It is also recommended that all patients with transient neurologic symptoms have a neuroimaging within 24 hours or as soon as possible.(Class 1,LOE B)

Diagnosis of Stroke

1. Imaging Studies

- **Non-contrast CT scan (NCCT):**
  - *First-line investigation to differentiate* ***ischemic vs. hemorrhagic stroke****.*
  - *Hemorrhagic stroke appears as* ***hyperdense (bright)*** *area.*
  - *Ischemic stroke appears as* ***hypodense (dark)*** *area after 24 hours.*
- **MRI (Diffusion-Weighted Imaging - DWI):**
  - *More sensitive for early ischemic changes.*
- **CT Angiography (CTA) or Magnetic Resonance Angiography (MRA):**
  - *Identifies large vessel occlusion or aneurysm.*

Stroke Diagnostic Tests

- Brain imaging: CT, MR
- Cardiac Imaging: TTE, TEE, heart monitoring
- Lipid, coagulation testing
- Vascular Imaging:

Noninvasive

- MR angiography (MRA)
  Intracranial, extracranial
- CT angiography (CTA)
  Intracranial, extracranial
- Ultrasound: Carotid, TCD

Invasive

- Conventional cerebral angiography

Differential Diagnosis of Stroke

- Ischemic stroke Hemorrhage stroke
- Craniocerebral / cervical trauma
- Meningitis/encephalitis
- Intracranial mass

•Tumor

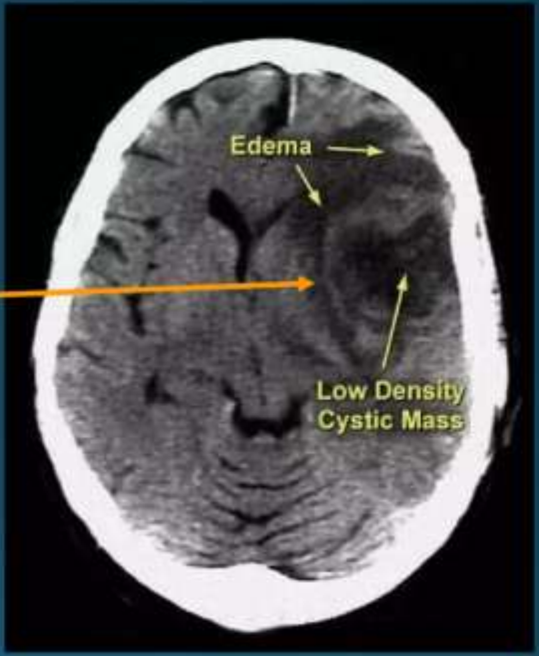
•Subdural hematoma

- Seizure with persistent neurological signs
- Migraine with persistent neurological signs
- Metabolic

•Hyperglycemia (nonketotic hyperosmolar coma)

•Hypoglycemia

•Post-cardiac arrest ischemia



Management of Stroke

1. **Ischemic Stroke Management**

- **Thrombolysis with IV Alteplase (tPA):**
  - *Given within **4.5 hours** of symptom onset.*
  - ***Contraindications:** Bleeding risk, recent surgery, uncontrolled hypertension.*
- **Endovascular Thrombectomy:**
  - *For large vessel occlusions, within **6-24 hours** of onset.*
- **Antiplatelet Therapy:**
  - ***Aspirin (150-300 mg/day) + Clopidogrel** for secondary prevention.*

Ischemic Stroke Management

- **Statins (e.g., Atorvastatin, Rosuvastatin 20-80 mg):**
  - *To reduce atherosclerosis risk.*
- Blood Pressure Control:
- Maintain BP < 180/105 mmHg during acute phase.
- Physiotherapy and Rehabilitation:
- Early mobilization prevents disability.

Hemorrhagic Stroke Management

- Blood Pressure Control: Target SBP <140 mmHg using IV Labetalol, Nicardipine.
- Surgical Intervention: Decompressive craniectomy for large hematomas.
- Aneurysm Clipping or Coiling in SAH.
- ICP Management: Mannitol or Hypertonic Saline for cerebral edema.
- Seizure Prophylaxis: In selected cases (e.g., lobar hemorrhage).

Prevention of Stroke

- Lifestyle Modification: Smoking cessation, diet control, exercise.
- Antihypertensive Therapy: ACE inhibitors, ARBs, or calcium channel blockers.
- Diabetes Control: HbA1c target <7%.
- Lipid Control: LDL target <70 mg/dL.
- Anticoagulation for Atrial Fibrillation: Warfarin or DOACs (Apixaban, Rivaroxaban).

POOR PROGNOSTIC FACTORS IN STROKE

- Accompanying fever
- Hypotension and severe hypertension
- Low oxygen saturation
- Hyperglycaemia and hypoglycemia
- Total anterior circulation stroke (55% dead)
- Pontine Haemorrhage
- Low GCS score
- heart failure
- severity of hemiparesis

Complications of Stroke

Acute

Hypertension 84%

Herniation – 78%

Cerebral oedema - 41%

Dysphagia – 51%

Hyperglycaemia – 28%

Arrhythmias – 31%

Seizures - 11%

SIADH – 10%

After 1st week/Chronic

Depression – 50%

DVT – 53%

Fever – 44%

Pneumonia – 35%

Septicaemia - 5%

Cardiac - 21%